# Multiplication Table Patterns

TEACHER KEY

3rd Grade Math · Interactive Math · Explore & Reflect

## Get started

Explore the 10 × 10 multiplication table. Highlight a row, shade even products, find every way to make a product, and solve missing-factor puzzles. Open the tool and use it to answer each question.

## Explore

**1.** Use the table: 8 × 9 = __

**72**

**2.** Highlight the 5s. What digits do the products end in?

**0 or 5**

**3.** Find every way to make 24 in the table.

**3 × 8, 4 × 6, 6 × 4, 8 × 3**

**4.** Why does the table look the same on both sides of the diagonal?

**a × b = b × a, so each product appears in two matching places.**

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## Explore more

**5.** Solve: 9 × __ = 63

**7**

**6.** Shade the even products. Is 4 times any number even or odd? Why?

**Even, because 4 × a number is double 2 × that number.**

## Reflect

**7.** Reflect: How can a pattern help you remember a hard fact like 9 × 6?

**For example, 9s products go up by 10 and down by 1 (54 is 60 − 6), and the digits add to 9.**

## Teacher notes

- Ask students to explain a pattern using properties: 4 × n is even because it is 2 × (2 × n).
- Use the puzzle button for unknown-factor practice (3.OA.A.4); each press shows a new puzzle.
- Arrow keys move the selected cell, which works well on a projector.

**Standards:** 3.OA.A.4, 3.OA.B.5, 3.OA.C.7, 3.OA.D.9. Accept any correct answer where several are possible, and ask students to explain with the tool's picture.